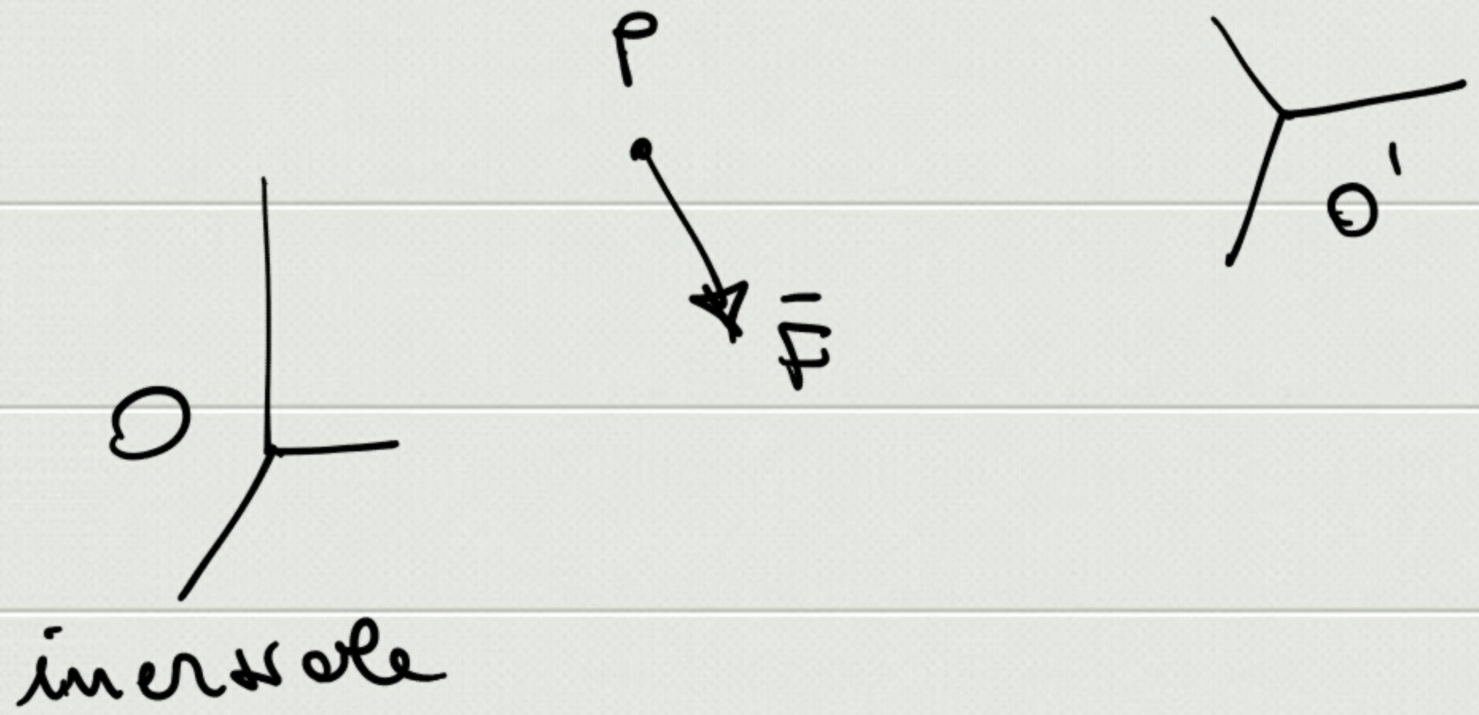


$\bar{v}_{O'} \neq 0$

$\bar{a}_{O'} \neq 0$

$\bar{\omega} \neq 0$

$$\bar{a} = \bar{a}' + \bar{a}_t + \bar{a}_c$$

$$\Rightarrow \boxed{\bar{a}' = \bar{a} - \bar{a}_t - \bar{a}_c}$$

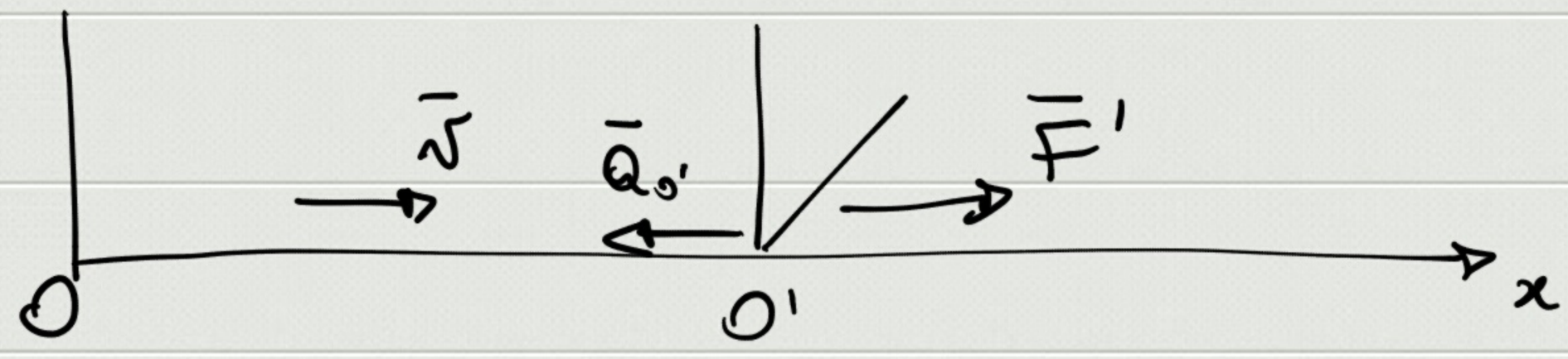
$$\bar{F} = m\bar{a}$$

$$\bar{F}' = m\bar{a}' = m\bar{a} - m\bar{a}_t - m\bar{a}_c =$$

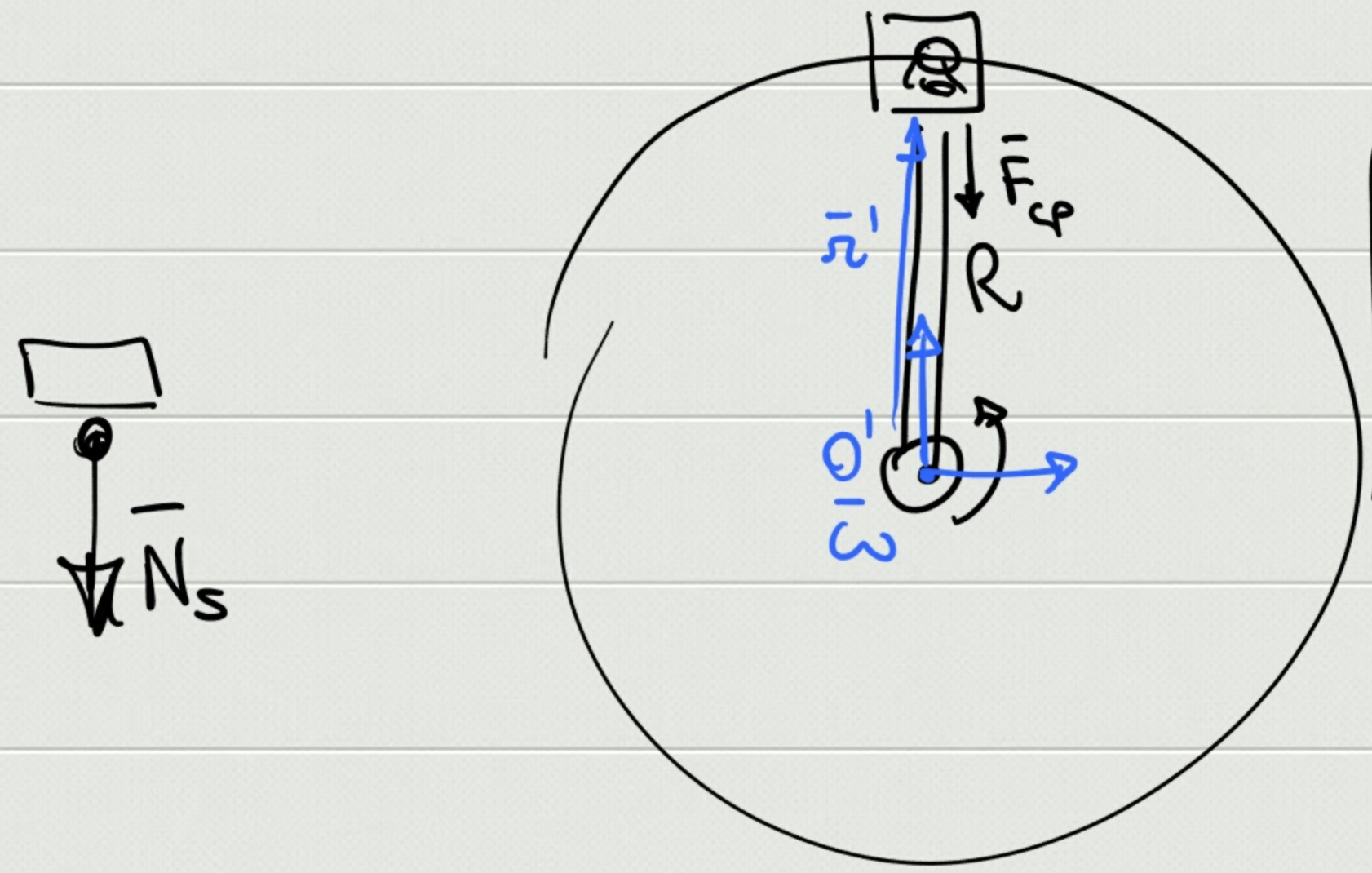
$$= \bar{F} + \underbrace{(-m\bar{a}_t - m\bar{a}_c)}_{\text{forze apparenti}}$$

$$\boxed{F=0} \Rightarrow a=0 \quad \Rightarrow \quad \bar{F}' = -m\bar{a}_t - m\bar{a}_c \neq 0$$

$$\overset{!}{=} m\bar{a}' \quad \Rightarrow \bar{a}' \neq 0$$



$$F' = -m\bar{a}_{O'}$$



$$\boxed{\begin{aligned} &\omega = \text{cost} \\ &v_{O'} = 0 \\ &\bar{a}_{O'} = 0 \end{aligned}}$$

$$\bar{F}_{cp} = N_s = m\omega^2 R \bar{u}_N$$

$\bar{r}' = \text{cost}$

$\bar{v}' = 0 \qquad \bar{a}' = 0$

$$\bar{F}' = m\bar{a} - m\bar{a}_t - m\bar{a}_c =$$

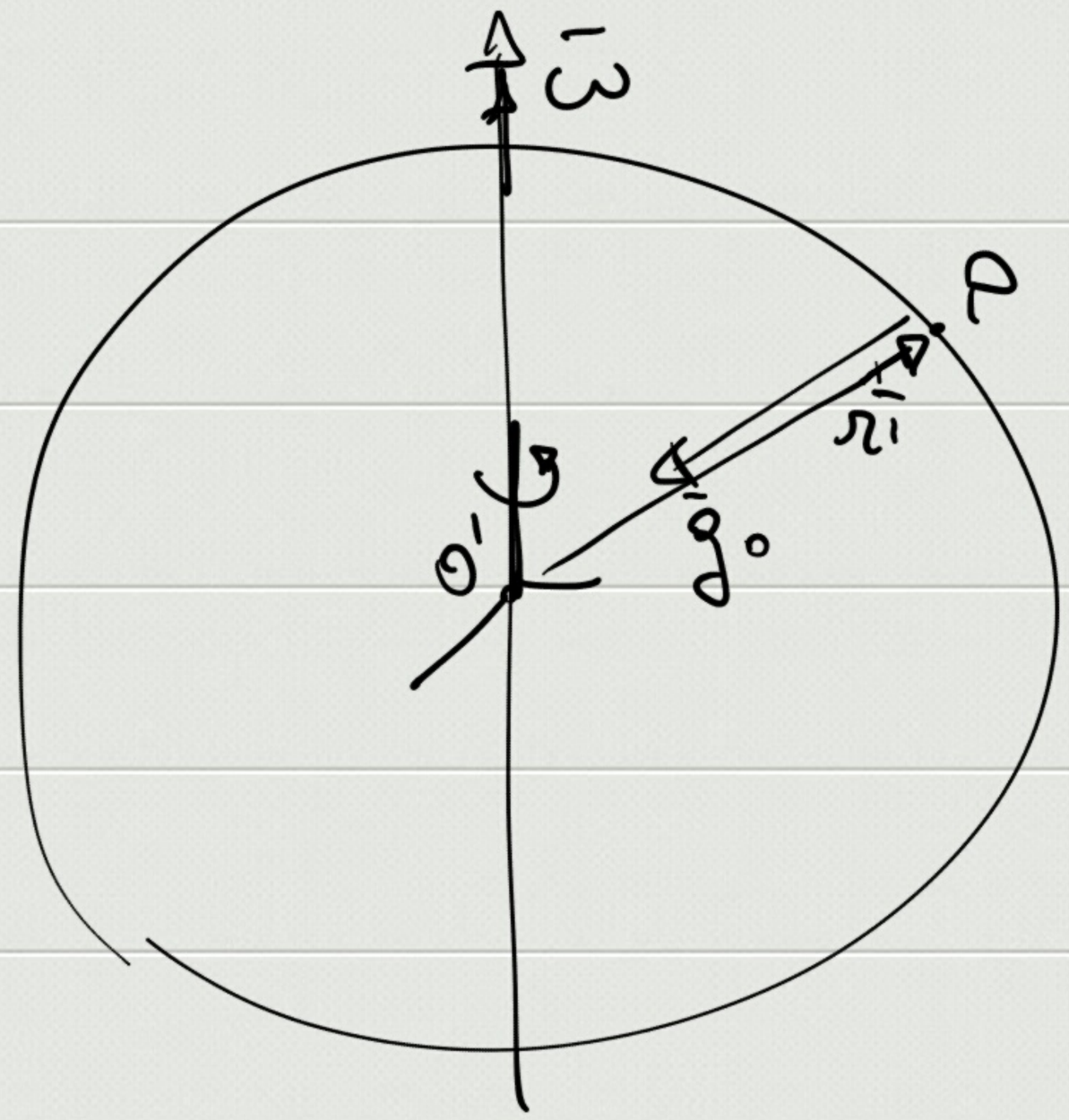
$$= m\omega^2 R \bar{u}_N - m\,\bar{\omega} \times (\bar{\omega} \times \bar{r}') = 0$$

$-m\bar{\omega} \times (\bar{\omega} \times \bar{r}')$

$[m\omega^2 R]$

modulo

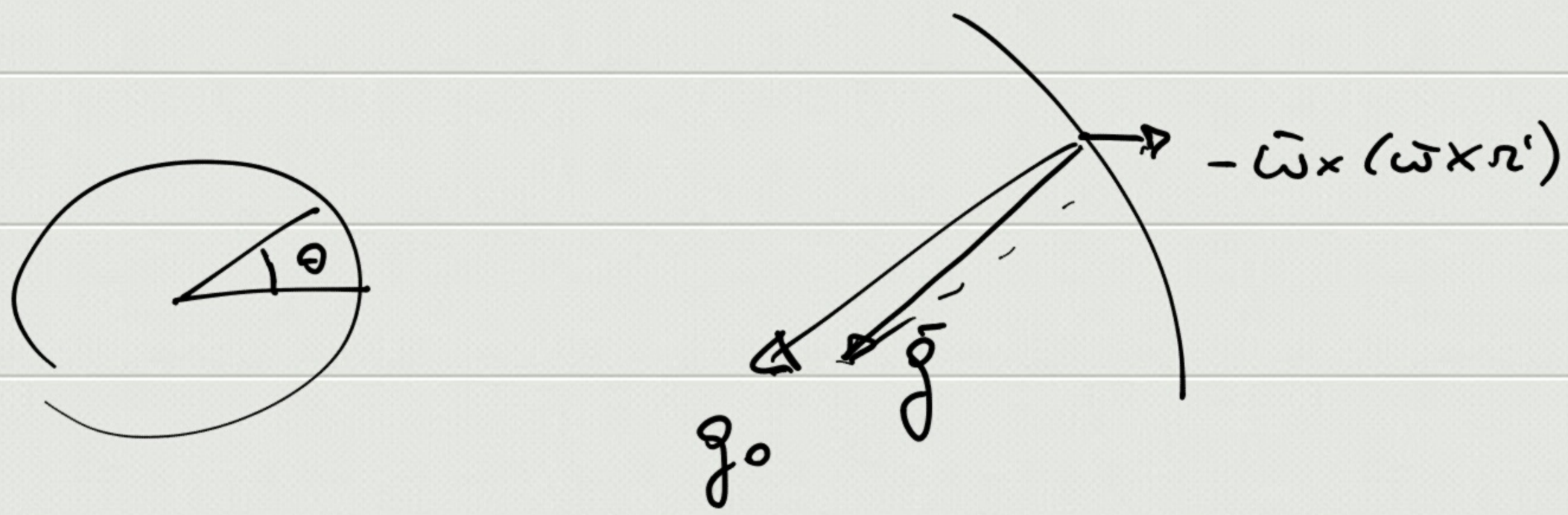
Forza centrifuga



$$\bar{g}_0 = \bar{g} + \bar{\omega} \times (\bar{\omega} \times \bar{r}') + 2\bar{\omega} \times \bar{v}'$$

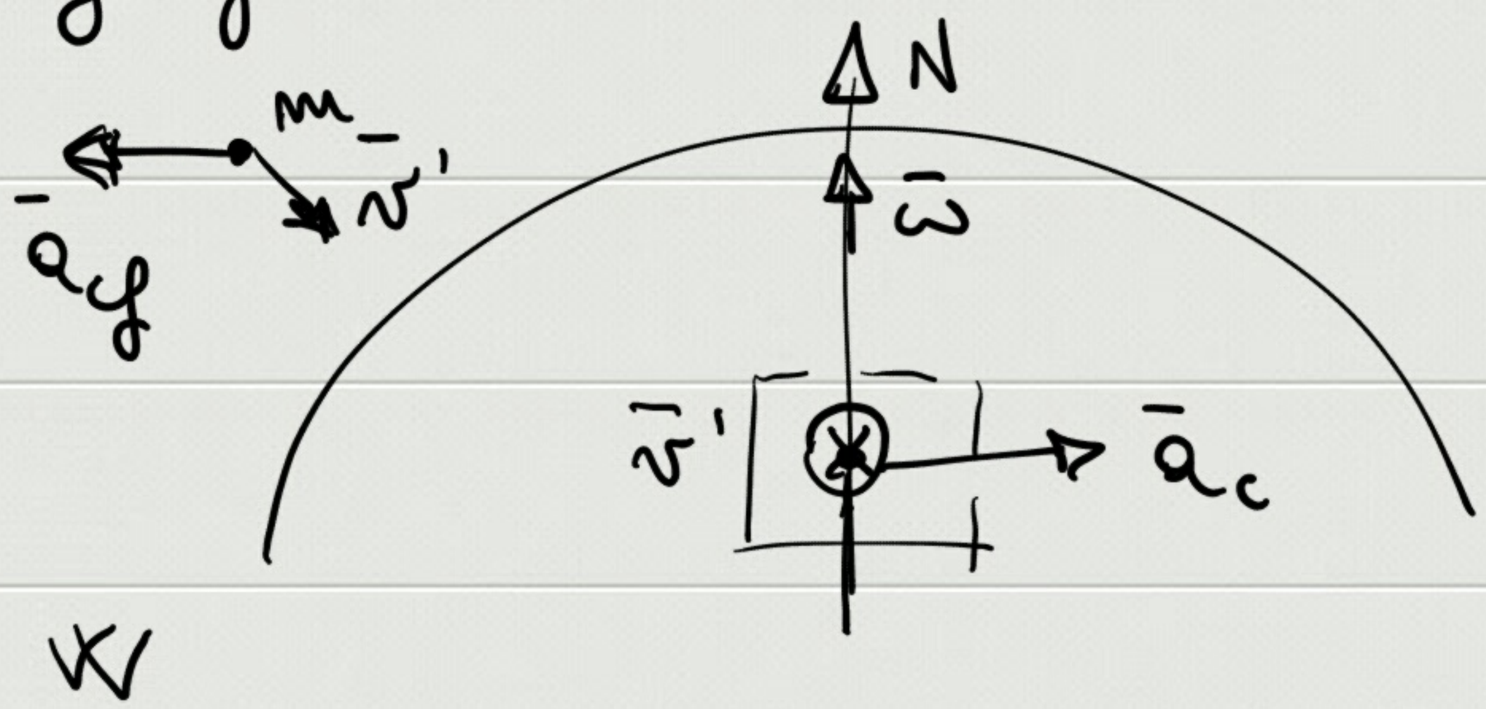
$$\bar{g} = \bar{g}_0 - \bar{\omega} \times (\bar{\omega} \times \bar{r}') - 2\bar{\omega} \times \bar{v}'$$

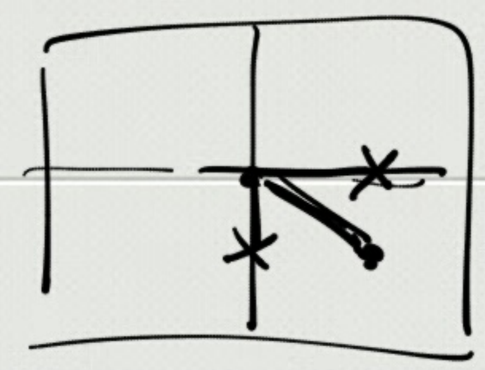
$\bar{v}' = 0$ $\qquad \bar{g} = \bar{g}_0 - \bar{\omega} \times (\bar{\omega} \times \bar{r}')$



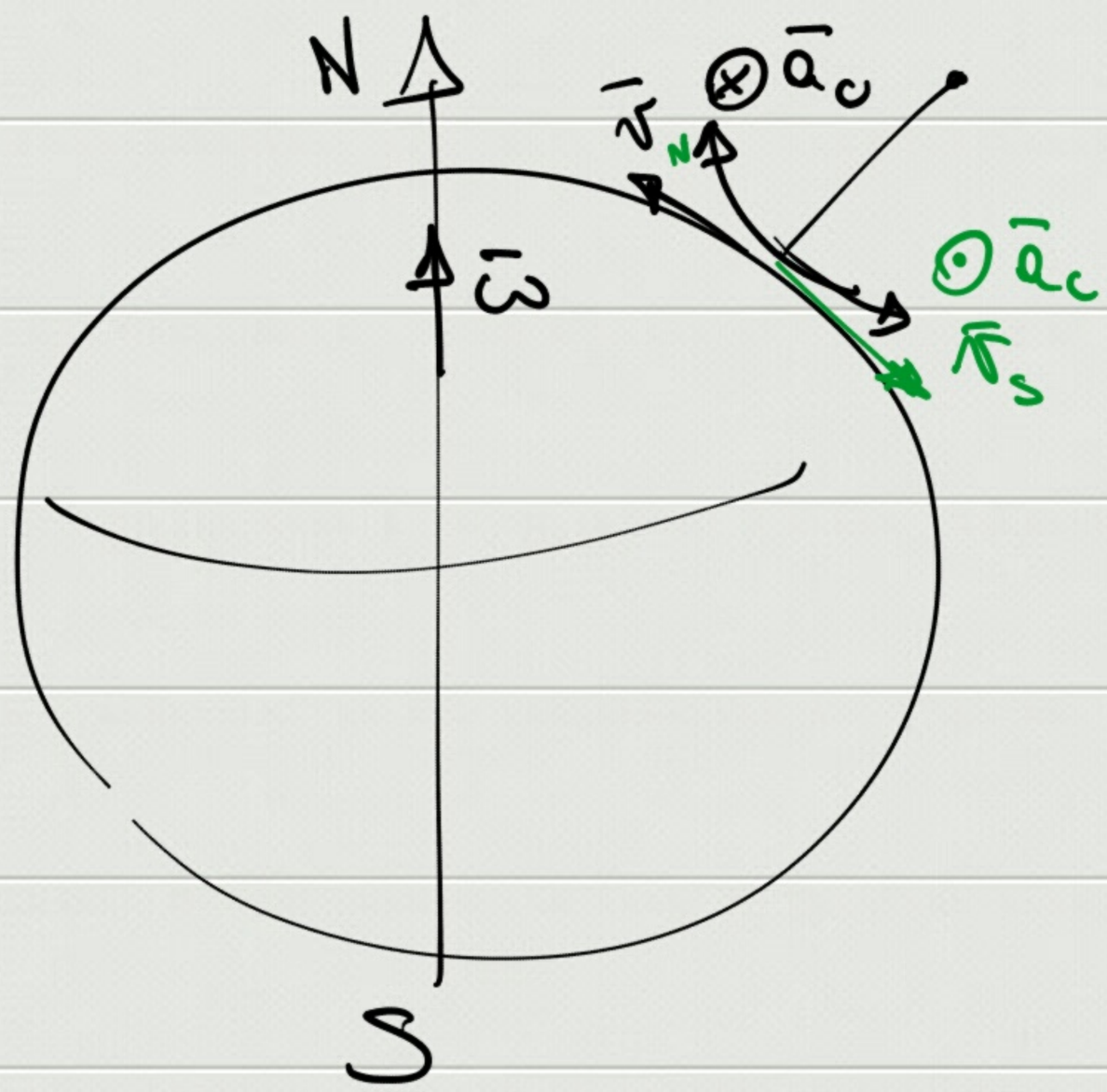
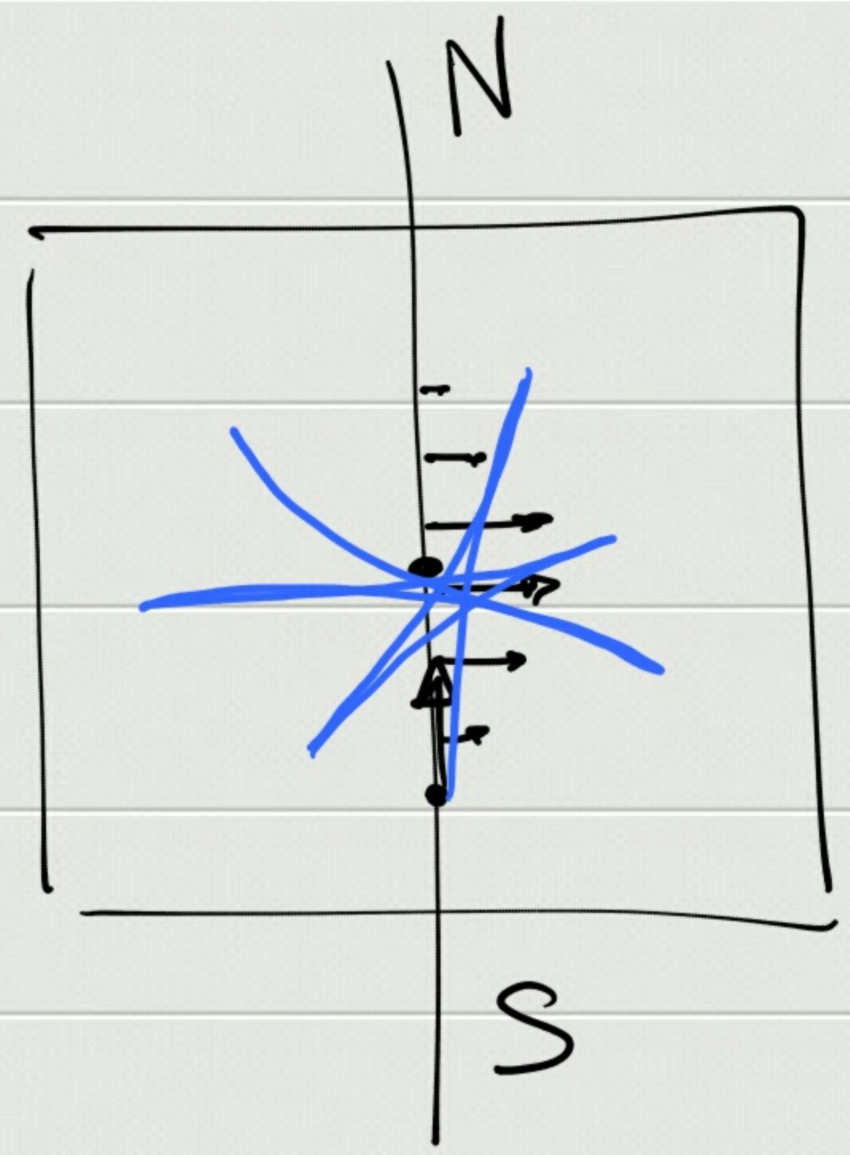
$$\omega^2 R_T \cos\theta = 3.38 \cdot 10^{-2} \cos\theta \quad m/s^2$$

$$\bar{g} = \bar{g}_0 - \bar{\omega} \times (\bar{\omega} \times \bar{r}') - 2\bar{\omega} \times \bar{v}'$$





$h = 100$ m $\quad \theta = 45°$

$\Delta x_{cf} = 17.3$ cm

$\Delta y_{cor} = 1.6$ cm